Technology Stack

Date	28 June 2026
Team ID	
Project Name	Voice Based Communication Understanding Analyzer
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	**Streamlit, HTML, CSS**	Streamlit provides a simple, interactive web interface for AI applications, while HTML/CSS improve the user interface with responsive layouts and custom styling.
2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	**Python 3.10+, Streamlit**	Python is well suited for AI, NLP, and audio processing. Streamlit simplifies backend integration and enables rapid application development.
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	**Local File Storage (Audio, Reports, Waveforms)** *(Future: SQLite/PostgreSQL)*	Uploaded audio files, generated transcripts, waveform images, and PDF reports are currently stored locally. Database integration can be added in future releases.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	**GitHub, Streamlit Community Cloud** *(Future: AWS/Azure)*	GitHub is used for version control, while Streamlit Community Cloud enables easy deployment. AWS or Azure can be adopted later for improved scalability.
5	Version Control & CI/CD (e.g., GitHub, GitLab, Jenkins)	**Git, GitHub**	Git provides source code version control, and GitHub enables collaboration, repository management, backup, and deployment integration.
6	Third-Party APIs / Other Tools (e.g.,	**OpenAI Whisper, Sentence-Transformers (Sentence-**	Whisper performs speech-to-text transcription, Sentence-BERT measures

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	Stripe, Twilio, Google Maps)	**BERT), Librosa, Matplotlib, ReportLab, PyTorch, NLTK, FFmpeg**	semantic similarity, Librosa extracts audio features, Matplotlib generates waveforms, ReportLab creates PDF reports, PyTorch supports AI models, NLTK provides NLP utilities, and FFmpeg handles audio processing.